CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Magnetic field lines emerge from the: 1

(a) north pole and enter at the south pole **(b)** south pole and enter at the north pole
(c) nowhere **(d)** middle of the magnet

2. The magnetic field inside a long current-carrying solenoid is: 1

(a) circular **(b)** uniform
(c) maximum at the ends **(d)** zero

3. The rule used to find the direction of force on a current-carrying conductor in a magnetic field is: 1

(a) right-hand thumb rule **(b)** Fleming's left-hand rule
(c) Snell's law **(d)** Ohm's law

4. A student states: “Two magnetic field lines never intersect.” The correct response is: 1

(a) Correct **(b)** Only near poles
(c) Incorrect **(d)** Cannot be determined

5. For the direction of the field around a straight wire, Student A uses the right-hand thumb rule; Student B uses Fleming's left-hand rule. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The colour of the insulation of the earth wire is usually: 1

(a) black **(b)** blue **(c)** green **(d)** red

7. The strength of the magnetic field at a point near a straight conductor increases when: 1

(a) the wire is cooled **(b)** the current decreases
(c) the distance increases **(d)** the current increases

8. The device that protects the circuit from overloading is: 1

(a) fuse **(b)** ammeter **(c)** switch **(d)** voltmeter

9. The magnetic field lines around a straight current-carrying conductor are: 1

(a) ellipses **(b)** concentric circles
(c) straight lines **(d)** parabolas

10. The force on a current-carrying conductor in a magnetic field is maximum when the conductor is: 1

(a) outside the field **(b)** perpendicular to the field
(c) at 45° **(d)** parallel to the field

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a magnetic field? 2

12. State the right-hand thumb rule. 2

13. What is short-circuiting? 2

14. A student says that the magnetic field lines are closed curves. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. State Fleming's left-hand rule. Name two factors on which the force on a conductor depends. 3

16. Describe the magnetic field produced by a current-carrying circular loop. 3

17. What is a solenoid? Draw (describe) its field lines. How can it be used to make an electromagnet? 3

OR

Why is earthing of appliances necessary? Which colour wire is used? 3

18. What is overloading? How can it be prevented? (two methods) 3

OR

How does the magnetic field around a straight wire change if 3
(i) current is doubled
(ii) distance from the wire is doubled
(iii) current direction is reversed?

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe an activity to show that a current-carrying conductor experiences a force in a magnetic field. 5
(ii) How does the direction of force change if the current is reversed?
(iii) Name the rule used to find the direction.

OR

(i) Draw (describe) the schematic diagram of a domestic electric circuit. 5
(ii) Why are two separate circuits used (15 A and 5 A)?
(iii) Role of the earth wire and fuse.

20. **Case study:** A student placed a compass near a straight wire carrying current upward. 5
(i) What happens to the needle?
(ii) What happens if the current is reversed?
(iii) What does it show?
(iv) Draw (describe) the field pattern.
(v) Name the rule for direction.

OR

(i) What are the properties of magnetic field lines? (any four) 5
(ii) How is the relative strength of a field shown by field lines?

END OF QUESTION PAPER — Total: 40 Marks